

Anderson–Rubin Test: Specification and Variable Construction

Companion to the May 14, 2026 slides

1 The estimating equation

Every cell on the slides comes from one cluster-robust OLS regression at the municipality–year level:

$$\log \text{GDP}_{mt} = \alpha_m + \delta_t + \gamma^c Z_{mt}^c + \psi^c \text{EC}_{mt}^c + \lambda \text{Vol}_{mt} + \eta_{mt}. \quad (1)$$

This is the AR *reduced form*: the GDP equation with the sector employment-share vector s_{mt} substituted out through its first stage on the instruments. The exposure controls EC_{mt}^c and the volume control Vol_{mt} are toggled on and off across the four control specifications of [section 2](#); the instruments Z_{mt}^c and the fixed effects α_m, δ_t enter every regression.

- Z_{mt}^c stacks the $K - 1$ retained sector instruments for alignment channel c ; EC_{mt}^c stacks the matching $K - 1$ exposure controls. Both are defined in [section 4](#).
- α_m, δ_t are municipality and year fixed effects. Standard errors are clustered by municipality.
- The reported **AR joint** F is the cluster-robust Wald statistic for $\gamma^c = \mathbf{0}$ —the joint significance of *channel c 's instruments only*. Under the optimality null $H_0 : \beta = \mathbf{0}$ this Wald test is the AR test. EC_{mt}^c and Vol_{mt} are partialled out, not tested.
- This is a *per-channel* run: each regression carries one channel's instruments, not a stacked four-channel set.

2 The four control specifications

The four columns of every slide table differ only in which controls join Z_{mt}^c on the right-hand side of (1).

Slide label	Regressors beyond FE	Purpose of the added control
No controls	Z^c	—
+ EC	Z^c, EC^c	absorb the baseline exposure share channel
+ Vol	Z^c, Vol	absorb the BNDES volume channel
+ Vol + EC	$Z^c, \text{EC}^c, \text{Vol}$	both

3 The four channels

Each alignment channel c pairs a set of political offices with an alignment shock Align_{mpt}^c . The cross-office shocks are products of tier-specific alignment indicators.

Channel	Offices	$\text{Align}_{mpt}^c = 1$ when party p ...
M	mayor	holds the mayoralty of m
M·P	mayor, president	holds the mayoralty <i>and</i> is in the presidential coalition
M·G	mayor, governor	holds the mayoralty <i>and</i> is in the governor's coalition of $s(m)$
M·G·P	mayor, gov., pres.	is aligned at all three tiers simultaneously

4 Variable construction

Outcome — $\log \text{GDP}_{mt}$

Municipal log GDP (at 2018 prices), 2002–2017.

Instrument — Z_{jmt}^c (**muni-relative owner share**)

1. **Primitive: owner-party counts.** $L_{f,p,s}$ is the number of owners of firm f affiliated with party p in year s . Owners with no party affiliation are dropped: the denominator counts politically-affiliated owner-years only.
2. **Pre-earliest-election window.** $T_t^c = [e_c^{\min}(t) - 4, e_c^{\min}(t) - 1] \cap [2002, 2017]$, where $e_c^{\min}(t)$ is the most recent election, on or before t , among the offices in channel c (mayoral elections 2000/04/08/12/16; coincident governor and president 2002/06/10/14). The window closes before any office in the channel can turn over, so the weights are predetermined relative to the shock.
3. **Owner-share weight.**

$$\tilde{w}_{jmp,t}^c = \frac{\bar{L}_{jmp,t}^c}{\bar{L}_{m,t}^{c,\text{affil}}} = \frac{\sum_{s \in T_t^c} \sum_{f \in \mathcal{F}(j,m)} L_{f,p,s}}{\sum_{j'} \sum_{p'} \sum_{s \in T_t^c} \sum_{f \in \mathcal{F}(j',m)} L_{f,p',s}},$$

the share of municipality m 's pre-window affiliated owner-years that sit in sector j at party p . The numerator $\bar{L}_{jmp,t}^c$ is the count of pre-window owner-years in cell (j, m) affiliated with party p — the firm-level counts $L_{f,p,s}$ of step 1 summed over the firms ever active in sector j of municipality m , $\mathcal{F}(j, m)$, and over the window years $s \in T_t^c$. The denominator $\bar{L}_{m,t}^{c,\text{affil}}$ sums that numerator over all sectors j' and parties p' in m . The muni-level normalization carries information about each cell's size in the local economy, which a within-cell intensity weight discards.

4. **Shift and stack.** $Z_{jmt}^c = \sum_p \tilde{w}_{jmp,t}^c \text{Align}_{mpt}^c$, then reshaped to one column per retained sector.

Exposure control — $\text{EC}_{jm,t}^c$

The same weights, summed over parties *without* the shift:

$$\text{EC}_{jm,t}^c = \sum_p \tilde{w}_{jmp,t}^c,$$

cell (j, m) 's share of the municipality's affiliated owner mass. It absorbs the share-component confounding of the shift-share instrument. One column per retained sector per channel.

Volume control — Vol_{mt}

$$\text{Vol}_{mt} = \frac{\text{BNDES_total}_{mt}}{\text{GDP}_{m,2002}},$$

represents the current-year total municipal BNDES disbursements over *initial-period* (2002) municipal GDP. The denominator is fixed pre-treatment, so Vol_{mt} varies in t only through the numerator.